

Problems 1 - 10 are Junior level 1.
 Problems 11 - 20 are Junior level 2, Senior level 1.
 Problems 21 - 30 are Junior level 3, Senior level 2.
 Problems 31 - 40 are Senior level 3.

The problems may appear in a different order in each section to the exam document - sorry about that!

Problem 1.

Which of the following computations yields the largest result?

- a) 20×23 b) $20 + 23$ c) $202 + 3$ d) 202×3 e) $20 \times 2 \times 3$

Answer: d)

Problem 2.

Which is the smallest odd positive integer which is neither a square nor a prime number?

- a) 1 b) 6 c) 9 d) 11 e) 15

Answer: e)

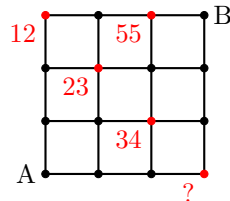
Problem 3.

A father is 41 years old and his daughter is 10 years old. In how many years is the father twice as old as the daughter?

Answer: 21

Problem 4.

A hundred people are moving along the sides of the graph below from A to B . They are only allowed to move upwards or to the right. The numbers indicate how many people passed through that checkpoint. How many people passed the checkpoint indicated with the question mark ?



- a) 21 b) 50 c) 17 d) 31 e) none

Answer: d)

Problem 5.

Yanta has drawn a snowflake on a piece of paper. She notices that there are multiple ways to put a one-sided mirror on the drawing, such that a part of the drawing and its mirror image together look just like the original drawing.

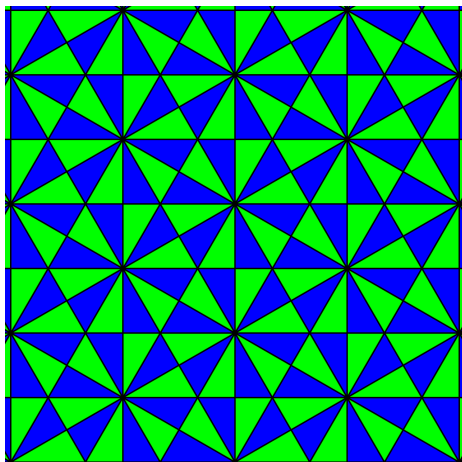
How many ways are there to put the mirror?

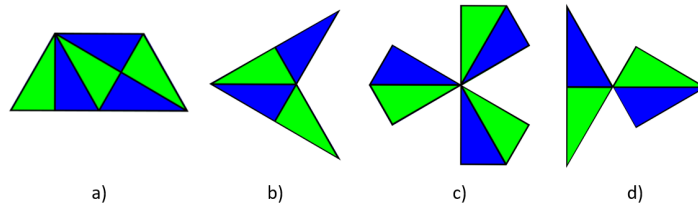


Answer: 12

Problem 6.

Which shapes appear in this tiling?

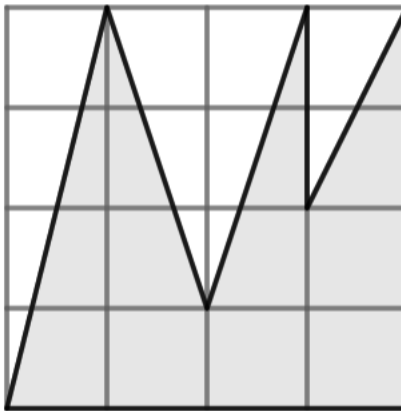




Answer: a), b), c)

Problem 7.

Given a square of side 20, we split every segment in 4 equal parts. What is the area of the grey region?



Answer: 250

Problem 8.

In a shop, there are a lot of candies in each of the colours red, yellow and green. All candy is either round or shaped like cube. Suppose that round candies are never green and yellow candies are always cube-shaped. Which of the following statements are necessarily true?

- A Red candies can be both shapes
- B Round candies are always red
- C Red candies are always round
- D Cube-shaped candies appear in at least 2 colours.

Answer: B,D

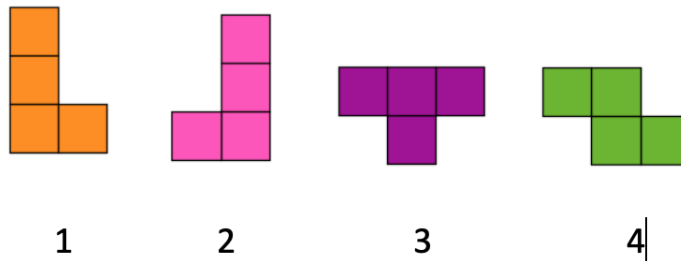
Problem 9.

At ETH, Professor Angst is in a lecture hall and counts the number of students. It's a two-digit number such that he can rearrange the digits to obtain a perfect square. What is the largest possible number of students in the hall?

Answer: 94

Problem 10.

Given the following 4 tetrominoes, Paul wants to create a 3×4 -rectangle using three different tetrominoes. Which one can he not use?

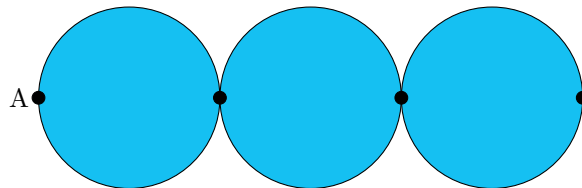


- a) Tetromino 1
- b) Tetromino 2
- c) Tetromino 3
- d) Tetromino 4
- e) It is not possible to build a 3×4 rectangle using three of these tetrominos.

Answer: c)

Problem 11.

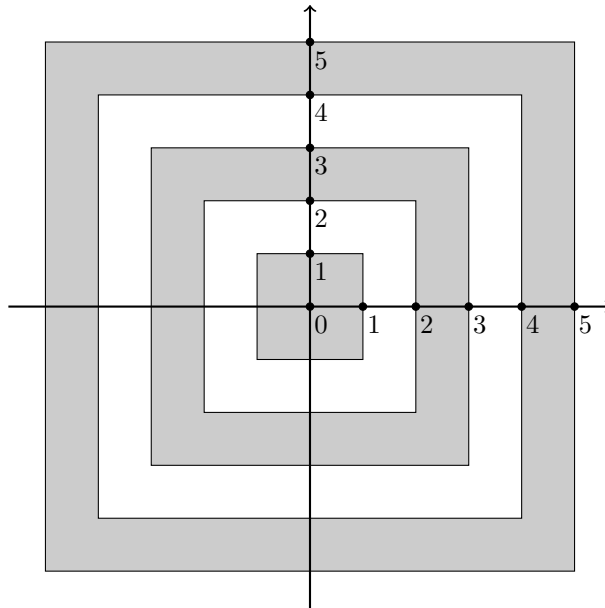
In a park, there are three round ponds. The shores of the ponds are partitioned into six segments in total, as depicted in the picture. Johann starts at point A and wants to walk along each of the six segments exactly once. How many different walks are possible?



Answer: 8

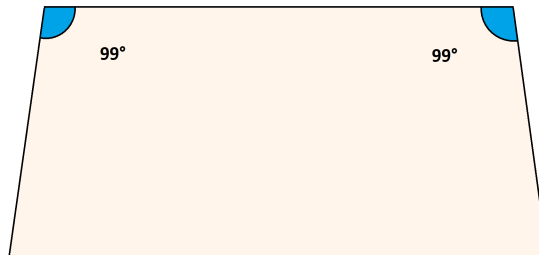
Problem 12.

All of the quadrilaterals in the picture below are squares. How big is the grey area?



Answer: 60

Problem 13.



In a big room there are many tables with the shape of an isosceles trapezoid. One of the two bigger angles is 99° . Viviane wants to put together some of the tables along their two shortest sides, so that the tables form a closed ring. How many tables does she need?

- a) 15 b) 18 c) 20 d) 24 e) 25

Answer: 20

Problem 14.

What is the smallest value that the following expression can take on, over all integers $x \geq 42$?

$$\frac{2023}{1 + \frac{1}{x}} + \frac{2023}{1 + x}$$

Answer: 2023

Problem 15.

In a 2×2 grid Annalena writes down the numbers 1, 2, 3, 4 in the four cells. She calculates the product for the two rows, the two columns and for the diagonal from top left to bottom right. She then sums up the five values and gets the number k . Which of the following isn't a possible value for k ?

- a) 23
b) 25
c) 27
d) 29
e) 31

Answer: b)

Problem 16.

Jana creates a burger which has four layers between the buns: patty, cheese, salad and a tomato slice. How many ways are there to arrange the four layers if the cheese has to be somewhere (not necessarily directly) above the patty?

- a) 3 b) 6 c) 8 d) 12 e) 16

Answer: 12

Problem 17.

The tap of Marco's bathtub is broken. Luckily, Marco has three buckets which measure 4, 5 and 16 litres respectively which he can use to fill the tub. What is the minimal number of times Marco has to fill up a bucket at the fountain to fill his tub with exactly 119 litres if he is not allowed to throw away any leftover water?

Answer: 11

Problem 18.

Valentin arranges 7 coins in a line, black on one and white on the other side. All the coins are black side up at the beginning. A move consists of Valentin picking a coin and flipping it and all the coins on its left. After precisely three moves, which of the following configurations are possible?

a) ●○○○○●●●

b) ○○●●●●●

c) ○●●●○●○

d) ○○●●○●●

Answer: b, d

Problem 19.

Anaëlle, Bibin, Clemens, David and Emily each got a paper with a number between 1 and 50. Their numbers are consecutive in some order. When comparing the numbers, they discover the following:

- Anaëlle: "My number is a prime number".
- Bibin: "Hey, mine too!".
- Clemens: "My number is right between Anaëlle's and Bibin's numbers, and it is divisible by 9".
- David: "My number is by 3 larger than Clemens' number"

What is Emily's number?

Answer: 20

Problem 20.

Six people participated in a chess tournament, where everyone played against everyone else exactly once. The winner of each match got 2 points and the loser got 0 points. In a draw, both players got 1 point. On the final scoreboard, there are five consecutive people with 2, 3, 4, 5 and 6 points respectively. How many points does the remaining person have?

- a) 0 b) 1 c) 7 d) 8 e) 10

Answer: e)

Problem 21.

Patrick forgot his 4-digit password, but he remembers the following properties:

- The two digit number formed by the first two digits, and the one formed by the last two digits are squares.
- The second and third digits are both prime numbers, and together they also form a two-digit prime number.

What is Patrick's password?

Answer: 2536

Problem 22.

In the following 6 boxes, the numbers 1 to 6 appear exactly once. What is the smallest possible value of the expression?

$$60 \cdot \left(\frac{\square}{\square} + \frac{\square}{\square} + \frac{\square}{\square} \right)$$

Answer: 69

Problem 23.

On a blackboard there is a four-digit number. The digits are strictly decreasing when read from left to right and the middle two digits are each strictly smaller than the average of their neighbouring digits. What is the largest possible number on the blackboard?

Answer: 9643

Problem 24.

There are 10 distinct lines given in the two-dimensional plane. How many intersection points can there be in total?

- A 0 B 1 C 3 D 45

Answer: A,B,D

Problem 25.

How many times during a day (24 hours) are the minute hand and the hour hand of a clock opposite from each other?

- a) 20 b) 22 c) 23 d) 24 e) 25

Answer: 22

Problem 26.

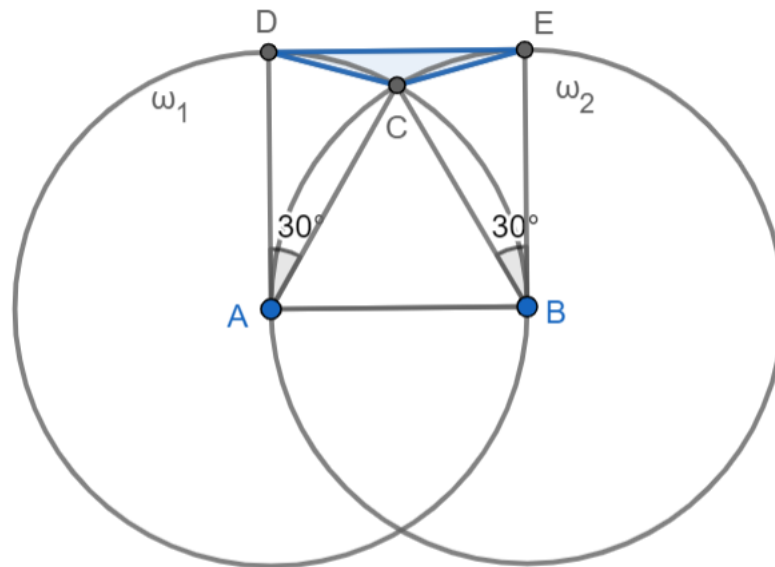
In a circle there are 2023 people. Each person is either a truth-teller or a liar. Each person in the circle says: "Both of their neighbours are liars!". Which of the following could be the number of truth-tellers?

- A 674
B 675
C 1011
D 1012

Answer: B, C

Problem 27.

Let ω_1 be a circle of centre A and radius 2. Let B be a point on ω_1 . Let ω_2 be a circle of centre B and radius 2. Let C be one of the intersections of both circles. Define D and E as the sketch presents. What is the area of triangle CDE ?



- a) $\pi-3$ b) $2 - \sqrt{3}$ c) $\frac{1}{3}$ d) $\frac{1}{2}$ e) $\frac{\pi}{2} - \sqrt{3}$

Answer: b

Problem 28.

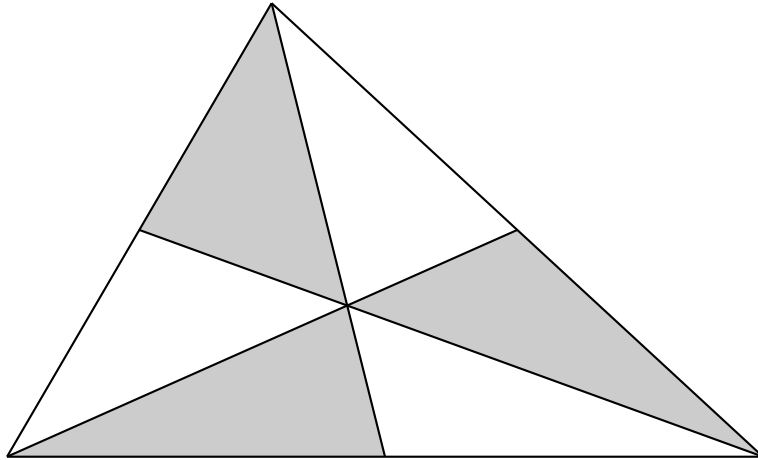
The number 1 is written on the blackboard. Matthew changes the number step by step. In each step he either multiplies it by 3 or he subtracts 1. What is the minimum number of steps needed to reach the number 2023?

- a) 10 b) 11 c) 12 d) 13 e) 14

Answer: 11

Problem 29.

Take an arbitrary triangle, which is divided by its medians into some regions. If the triangle has area 1, how big can the grey area be at most? *A median is a line connecting a vertex of the triangle to the midpoint of the opposite segment.*



If the triangle has area 1, how big can the grey region be at most?

- a) $\frac{1}{3}$ b) $\frac{1}{2}$ c) $\frac{3}{5}$ d) $\frac{2}{3}$ e) $\frac{3}{4}$

Answer: b)

Problem 30.

In a 4×4 grid, Henning wrote a number on each cell. Then, he observed that the sum of numbers in each column, in each row and in both main diagonals are the same. Call this number the magic sum. Unfortunately, Tanish was naughty and decided to erase some of the numbers, which leads to the displayed square. What is the magic sum?

7	27	29	
17		11	
9	21	19	
			25

Answer: 64

Problem 31.

A circle is divided into 5 sections. How many ways are there to colour the sections using three different colours such that no two adjacent sections are of the same colour?

Answer: 30

Problem 32.

How many three-digit numbers are divisible by their leftmost digit?

Answer: 286

Problem 33.

We fill each cell of a 10×10 table with $+1$ or -1 . What's the biggest possible number k such that there are exactly k rows with sum strictly greater than 0 and k columns with sum strictly smaller than 0?

- a) 5 b) 6 c) 7 d) 8 e) 9

Answer: d)

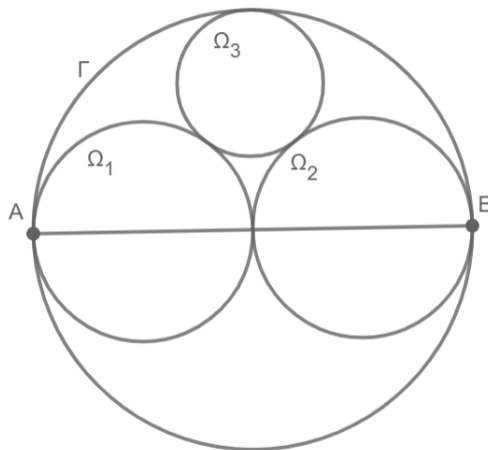
Problem 34.

There are 1000 suspicious people standing in a line. One of them has hidden a diamond in their pocket and all 1000 people know who. The police will ask everyone: "How many people are standing between you and the person with the diamond?". Luckily, the police knows that at least k people will answer truthfully. What is the minimal number of k needed in order for the police to find the diamond with certainty?

Answer: 502

Problem 35.

Given a circle Γ of radius 6 and AB be a diameter. Consider two small circles Ω_1 and Ω_2 of radius 3 such that A belongs to Ω_1 , and B to Ω_2 . All 3 circles are pairwise tangent. Let Ω_3 be tangent to all 3 other circles. What is the radius of Ω_3 ?



- a) $\pi/2$ b) 2 c) $\frac{\sqrt{3}+1}{2}$ d) $6 - \sqrt{41}$ e) 1

Answer: b)

Problem 36.

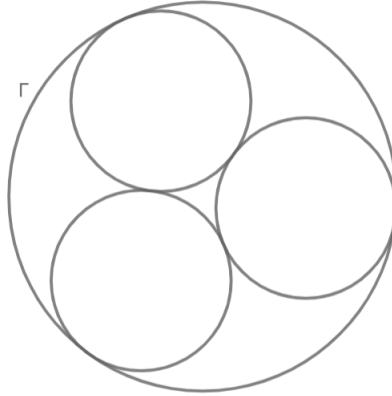
Julia wrote down the number 6 on her blackboard. She can now repeatedly replace the current number n either with n^2 or $n - 4$. Which numbers can she eventually arrive at?

- A 32
B -2022
C 500
D 2022

Answer: A, B, C

Problem 37.

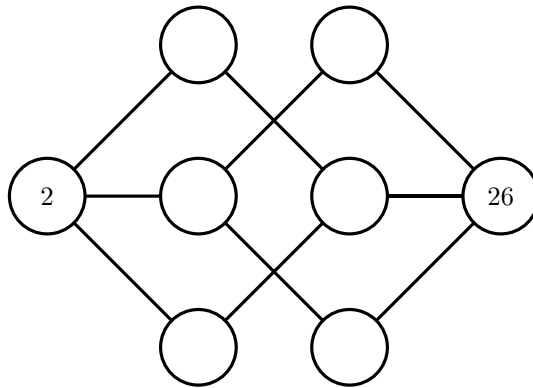
Let Γ be a circle of radius 1 consider 3 circles of radius r tangent to Γ and pairwise tangent. What is r ?



- a) $\frac{\sqrt{3}}{2+\sqrt{3}}$ b) $\frac{2}{7}$ c) $\frac{\pi}{8}$ d) $\frac{\sqrt{3}}{3}$ e) 0.4 *Answer:*
 $\frac{\sqrt{3}}{2+\sqrt{3}}$

Problem 38.

On a paper there are 8 circles, some of them are connected (see picture below). Two circles already contain the numbers 2 and 26. Nicole writes a number in each of the six remaining circles such that in the end, all numbers she wrote are the average of the numbers in the neighbouring circles. (Two circles are neighbouring if they are connected by a segment). What is the sum of all the eight numbers?



Answer: 112

Problem 39.

Consider the 25 two-digit numbers which only contain the digits 1, 2, 3, 4 and

5. Noah wants to place some of them along the boundary of a circle, such that the last digit of each number is equal to the first digit of the next number in clockwise direction. If he cannot use the same number twice, how many numbers can Noah place at most?

- a) 25 b) 24 c) 21 d) 20 e) 19

Answer: a)

Problem 40.

We call a natural number n *amazing* if it has at least 4 different divisors and if the sum of its four largest divisors is exactly equal to $2n$. Which of the following statements is true about amazing numbers?

- A There exist less than 100 amazing numbers.
B All amazing numbers are divisible by 3.
C There exists an amazing number ending with the two digits 12.
D There exists an amazing number ending with the two digits 22.

Answer: B, D